

Grade 7

Unit 1

Lesson 1 Practice

Name Answer Key

Date _____

Period _____

For questions 1-4, identify whether each question is statistical or not statistical. If it is not, rewrite the question so that it can be statistical.

1. How many days are in January?

Not statistical.

Answers may vary.

How many hours did you read each day in January?

2. What percent of students at school take the bus?

Statistical

3. How old is your cat?

Not statistical.

Answers may vary.

How old are all the cats in your neighborhood?

4. What is the most popular favorite flavor of ice cream amongst 7th graders?

Statistical

5. What color buttons are in the jar?

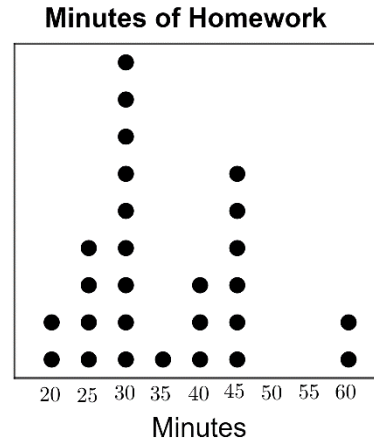
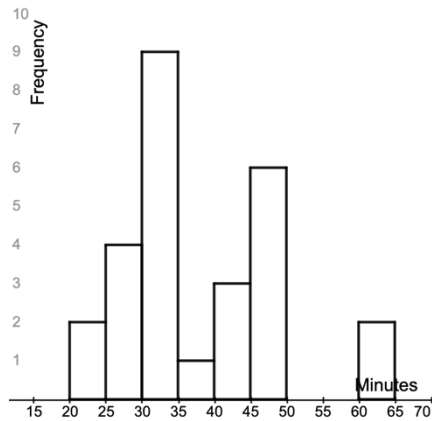
Not statistical.

Answers may vary.

What is the diameter of each red button in the jar?

For questions 6-10, use the scenario.

Teachers at Sanford Middle School were collecting data to determine approximately how much homework 7th graders received daily. They made a histogram and a line plot using the data they collected from surveying a sample of students.



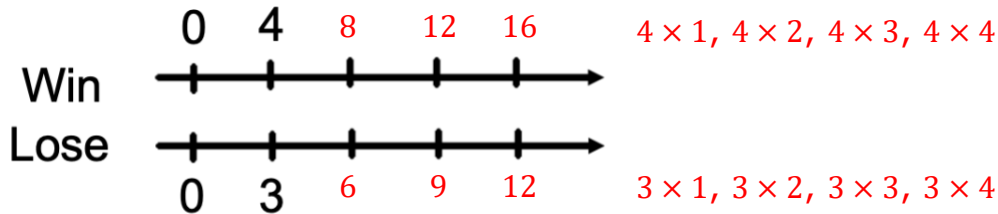
6. Which graph can be used to determine how many students were surveyed?
Either/Both
7. How many students were surveyed?
27
8. What percentage of students reported spending 35 minutes or less on homework every night? Round to the nearest tenth.
 $\frac{16}{27} \approx 59.3\%$
9. What is the range of minutes spent on homework?
 $60 - 20 = 40$
10. Are there any outliers? If so, name them.
No outliers.

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Unit 1
Lesson 2 Practice

Name Answer Key
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The Tampa Bay Thunder high school soccer team recently finished their spring season. The ratio of wins to losses during the season was 4:3.

1. Complete the double number line below using the ratio.



2. If the team lost 9 games, how many did they win?

$$\frac{12 \text{ wins}}{9 \text{ losses}} = 12 \text{ wins}$$

3. If the team played 35 games in the season, how many did they lose?

$$\frac{3 \text{ losses}}{7 \text{ total}} \times 5 = \frac{?}{35 \text{ total}} \quad 3 \times 5 = 15 \text{ losses}$$

Three friends decided to take a road trip from Jacksonville to New Orleans to see a football game. They split up the drive amongst them and switched drivers every time they stopped for gas or food. Their driving times and distances are recorded in the table below.

4. Complete the table to determine the unit rate for each friend. Round to the nearest tenth is needed.

Name	Distance (miles)	Time (hours)	Unit Rate (miles per hr.)
Mitch	202	3.5	$\frac{202}{3.5} = 57.7$
Timothy	162	3	$\frac{162}{3} = 54$
Jackson	147	2.5	$\frac{147}{2.5} = 58.8$

5. Who was driving the fastest?
Jackson at 58.8 miles per hour

6. If Timothy drove the entire way back at the same speed he drove on the way to New Orleans, how long would it take to get back? Round to the nearest tenth, if needed.
 $202 + 162 + 147 = 511$ total miles
 $511 \text{ miles} \div 54 \text{ miles per hour} \approx 9.5$ hours

An electronic store is having a sale where all items are 70% of their original price.

7. If all items are 70% of their original price, what is the percent discount will customers receive?
 $100 - 70 = 30$ They will receive a 30% discount

8. A new 4K TV originally costs \$2500. Complete the table.

	Price(\$)	Percentage
$2500 \times 0.10 = 250$	250	10
$2500 \times 0.30 = 750$	750	30
$2500 \times 0.70 = 1,750$	1,750	70
$2500 \times 1.00 = 2,500$	2,500	100

9. What is the amount of discount on the TV?
\$750 off (30% of 2500 is 750)

10. What is the sale price of the TV?
\$1,750 (70% of 2500 is 1750)

For questions 1-4, use the five rational values shown.

$$-1\frac{3}{4}, \quad 57\%, \quad 0.624, \quad -2.2, \quad 1.95$$

1. Which value is the smallest?
Make them all decimals or all fractions: $-1.75, 0.57, 0.624, -2.2, 1.95$

Smallest = -2.2

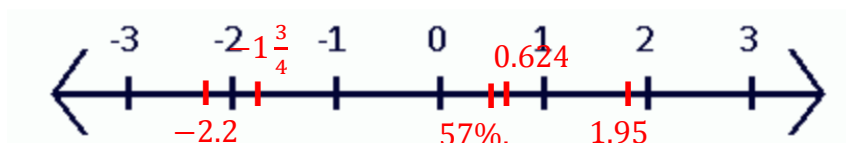
2. Which value is the largest?
Order from smallest to largest: $-2.2, -1.75, 0.57, 0.624, 1.95$

Largest = 1.95

3. Write each number as a ratio of two integers.

Number	Ratio of Two Integers
$-1\frac{3}{4}$	$-\frac{7}{4}$
57%	$\frac{57}{100}$
0.624	$\frac{624}{1000}$ or $\frac{78}{125}$
-2.2	$-\frac{22}{10}$ or $-\frac{11}{5}$
1.95	$\frac{195}{100}$ or $\frac{39}{20}$

4. Plot each value on the number line. Label each point with its numeric value.



5. True or False? *The absolute value of a number is always the opposite value.*

False

6. The distance a number is from zero on the number line is called the
absolute value.

For questions 7-10, use $<$, $>$, or $=$ to compare each pair of values.

7. $|-13|$ $>$ -14.6

$$13 > -14.6$$

8. $|-26\frac{1}{4}|$ $=$ $|26.25|$

$$26\frac{1}{4} = 26.25$$

9. 15.31 $>$ $|9.75|$

$$15.31 > 9.75$$

10. Identify the value(s) that make the statement below true.

$$| \underline{8.32} | = 8.32$$

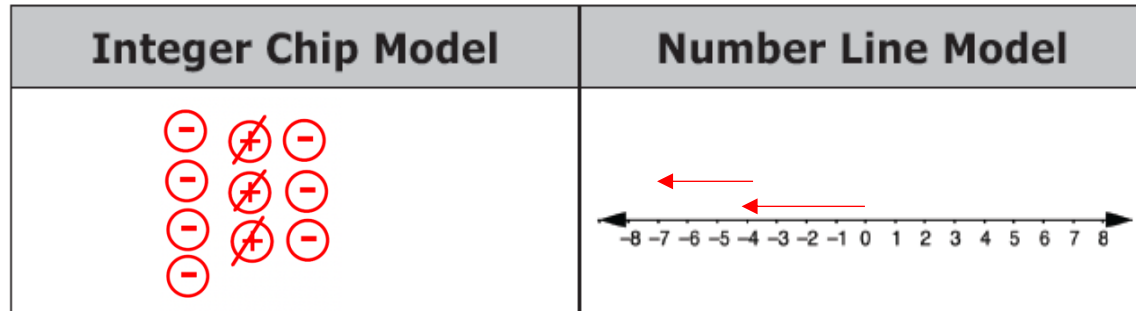
and
 $|-8.32|$

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Lesson 4 Practice

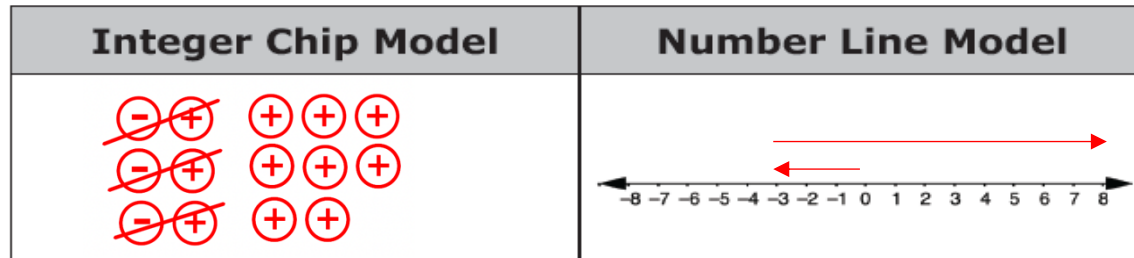
Name Answer Key
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Model each addition or subtraction using the integer chip model and the number line model.

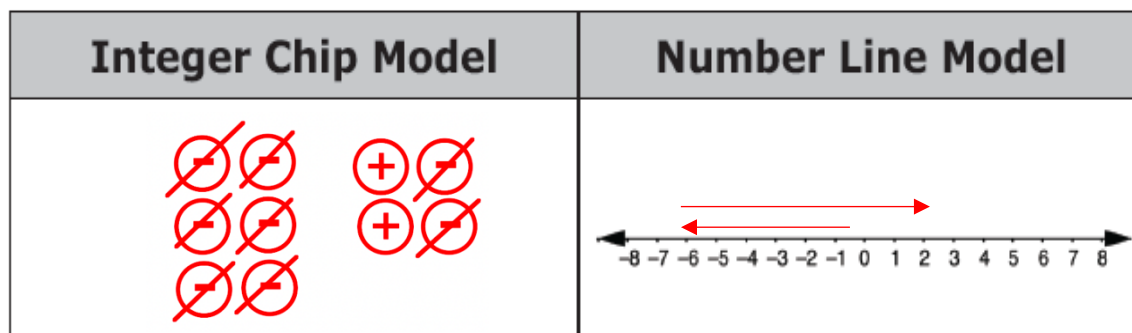
1. $-4 - 3 = \underline{-7}$



2. $-3 + 11 = \underline{8}$



3. $-6 - (-8) = \underline{2}$



4. When multiplying or dividing integers with the **same** sign, the sign of product or quotient will be

- ☐ positive
☐ negative

5. When multiplying or dividing integers with **opposite** signs, the sign of product or quotient will be

- ☐ positive
☐ negative

Evaluate each expression.

6. $-156 + (-97) = -253$

$$\begin{array}{r} 156 \\ + 97 \\ \hline 253 \end{array}$$

7. $63 \div (-9) = -7$

Opposite signs = -

8. $-8 \cdot (-7) = 56$

Same signs = +

9. $-34 - (-17)$

$-34 + 17$

Find the difference, keep the sign of larger absolute value

$$\begin{array}{r} 34 \\ -17 \\ \hline 17 \end{array}$$

10. $-6 \times 5 = -30$

Different signs = -

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Lesson 5 Practice

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1. In order to solve the equation $-12 = x + 7$, the equation is -19.

- ☐ add 7.
- ☒ subtract 7.
- ☐ multiply by 7.
- ☐ divide by 7.

The solution to

2. In order to solve the equation $\frac{y}{-3} = -5$, the equation is 15.

- ☐ add -3.
- ☐ subtract -3.
- ☒ multiply by -3.
- ☐ divide by -3.

The solution to the

3. In order to solve the equation $-55 = 11b$, the equation is -5.

- ☐ add 11.
- ☐ subtract 11.
- ☐ multiply by 11.
- ☒ divide by 11.

The solution to

4. In order to solve the equation $a - 17 = -9$, the equation is 8.

- ☒ add 17.
- ☐ subtract 17.
- ☐ multiply by 17.
- ☐ divide by 17.

The solution to

For questions 5 - 10, solve each equation using inverse operations.

5. $\frac{-2c}{-2} = \frac{-18}{-2}$ 9

$c = 9$

6. $6 = -10 + c$ 16
 $+10 \quad +10$

$16 = c$

7. $c - 8 = -23$ -15
 $+8 \quad +8$

$c = -15$

8. $-7 \cdot 6 = \frac{c}{-7} \cdot -7$ -42

$-42 = c$

9. $7 = c + (-3)$ 10
 $+3 \quad +3$

$10 = c$

10. $\frac{6y}{6} = \frac{-54}{6}$ -9

$y = -9$

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Unit 1
Lesson 6 Practice

Name Answer Key
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Period _____

Quinn went shopping at her favorite store. She spent \$96 on four shirts that each cost the same amount. What is the cost of each pair of each shirt?

1. Write an algebraic equation to represent the relationship.

$$4s = 96$$

2. What does the variable in your equation represent?

s = the cost of one shirt

3. Solve the equation.

$$\begin{array}{r} 4s = 96 \\ \underline{4} \quad \underline{4} \\ s = \$24 \end{array}$$

The students at Paul's middle school want to make \$350 in a bake sale to raise money for an upcoming trip. They have already made \$98. How much more money must they make to reach their goal?

4. Write an algebraic equation to represent the relationship.

$$98 + m = 350$$

5. What does the variable in your equation represent?

m = how much more money they need to raise to meet their goal

6. Solve the equation.

$$\begin{array}{r} 98 + m = 350 \\ -98 \quad -98 \\ m = \$252 \end{array}$$

For questions 7 - 8, use the equation $x + 5 = 24$.

7. Create a real-world problem that the equation could represent.

Answers may vary.

Chris wants to have 24 marbles. He already has 5 marbles. How many more marbles does he need to collect?

8. Identify what the variable represents in your scenario.

x = how many more marbles he needs to collect

For questions 9 - 10, use the equation $\frac{b}{7} = 3$.

9. Create a real-world problem that the equation could represent.

Answers may vary.

People signed up in a soccer league. They were placed into 3 teams of 7 people.

10. Identify what the variable represents in your scenario.

b = the total number of people in the soccer league